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The Upper Kaena Sedimentary Geomagnetic Reversal Record From Southern Sicily

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A detailed paleomagnetic and rock magnetic record of the upper Kaena polarity reversal was determined from late Pliocene marine marls in southern Sicily. Two separate characteristic magnetization components were distinguished: a low-temperature (LT) component between 200° and 500°C and a high-temperature (HT) component between 500° and 580°-610°C. The two components record the reversal in a slightly different way, but both were acquired in an early stage following deposition, most probably by authigenic (biogenic) formation of magnetic minerals. The actual (R-N) reversal, via E and horizontal inclinations, seems to take 3.3 kyr and is followed by a stable normal interval (3.3 kyr) and subsequently by a significant excursion (50°-60°) in declination (3.6 kyr). Relative intensities were lower during a time-span of 5.6 kyr and show a minimum of 10% of the nontransitional values, but during the excursion, no intensity changes are apparent. The VGPs of the HT component are confined to a longitude between India and Australia; those of the LT component oscillate between this same longitude and one passing over North and South America. We conclude that most if not all observed features are largely determined by the remanence acquisition process which is complicated by authigenic formation of magnetic minerals and cyclically fluctuating paleoredox conditions. This has not only resulted in considerable smoothing but also in delayed NRM acquisition in specific sedimentary intervals. We believe that the observed "excursion" is a sedimentary artifact and not related to geomagnetic field behavior during transitions. We emphasize that many reported "geomagnetic" features may in fact be due to the sedimentary remanence acquisition process.

INTRODUCTION

Detailed records of geomagnetic reversals may be obtained from different sources, most important of which are those recorded in lava sequences and those in sedimentary sequences. Both recording media have their advantages and drawbacks. The main advantages of sedimentary sequences concern appropriate time control and continuous registration of the geomagnetic signal. In addition, long sedimentary sequences from the same area may yield useful information concerning the characteristics of a number of successive polarity reversals [*Valet et al.*, 1988]. However, a major drawback of sedimentary records is the insufficient understanding of the acquisition of the remanence which may involve complex processes of authigenic formation of different magnetic minerals at different depths in the sediment during accumulation [e.g., *Lund and Karlin*, 1990; *Karlin*, 1990*a,b*; *Stolz et al.*, 1990; *van Hoof and Langereis*, 1991].

Long-term magnetostratigraphic research in the Central Mediterranean by our laboratory has so far resulted in the identification and location of 25 successive reversals from 7.5 to 1.5 Ma in the upper Miocene of Crete [*Langereis*, 1984] and in the Pliocene of Sicily and Calabria [*Zijderveld et al.*, 1986; *Zachariasse et al.*, 1989, 1990; *Langereis and Hilgen*, 1991]. A succession of three older reversals (11-12 Ma) from the same area is known from Zakinthos [*Laj et al.*, 1988]. Not all reversals are suitable for detailed paleomagnetic research, for exemple, due to low intensities and/or unfavorable rock magnetic properties. From the nine upper Miocene reversals, five were extensively investigated earlier [see *Valet et al.*, 1988, and references therein]. From the lower Pliocene in Calabria, five successive reversal records (upper Thvera, upper and lower Sidufjall and upper and lower Nunivak in the Gilbert Chron) have been reported by *Linssen* [1988, 1991]. From the upper Pliocene in Sicily, three more successive reversal records (upper and lower Mammoth and lower Kaena in the Gauss Chron) were also studied in detail by *Linssen* [1991]. This paper presents the details of the fourth successive reversal, the upper Kaena reversal, from the same sedimentary sequence.

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GEOLOGICAL SETTING AND SAMPLING

The Mammoth and Kaena reversals in the Gauss Chron were unambiguously identified by a detailed magnetostratigraphic study of the Punta Piccola section in the Caltanisetta basin of southern Sicily (Figure 1) [*Zachariasse et al.*, 1989, 1990]. The Punta Piccola section has a bedding plane with strike and dip of 315° and 11° NE and consists of marine marls of the Pliocene Trubi formation in which the Mammoth subchronozone is located and of the very similar Pliocene Monte Narbone formation in which the Kaena subchronozone is located (Figure 2). The marls of the Trubi and Monte Narbone formation consist mainly of carbonates (50-80% $CaCO_3$) and a mixture of clay minerals [*Hilgen*, 1987; *de Visser et al.*, 1989; *Linssen*, 1991]. The average sedimentation rate per polarity zone in the Punta Piccola section can accurately be determined and is remarkedly constant: 4-5 cm/kyr; the sedimentation rate in the Kaena polarity subzone is 4.3 cm/kyr [*Zachariasse et al.*, 1989].

Hilgen [1987] recognized a long succession of small-scale sedimentary cycles, so-called quadruplets. These quadruplets form a repetition of grey, white, beige, and white coloured beds (Figure 2) and were deposited during cyclic sedimentation periods which are related to the *Milankovitch* [1930] cycle of precession of the Earth's orbit [*Hilgen and Langereis*, 1989] with

an average duration of 21.7 kyr [*Berger and Loutre*, 1988]. The low-carbonate grey marls probably represent a period of increased continental runoff and a mainly suboxic environment [*de Visser et al.*, 1989]. The high-carbonate white marls represent a more oxic environment, but only those from the southwestern part [sicily, southern Calabria] of the Trubi formation contain the intercalated low-carbonate beige interval [*Hilgen*, 1987]. This beige interval is probably due to increased wind-blown African input [*de Visser et al.*, 1989]. In some of the grey marls from the Monte Narbone formation a dark brown and sometimes finely laminated sapropelitic layer is intercalated [*Hilgen*, 1987], as is the case in the sampled upper Kaena record [Figure 2). Although the weathering profile shows quite sharp changes in color and induration, the changes in fresh, unweathered sediment are much more gradual: light blue (white layer] to dark blue-grey (grey layer). The color boundaries between grey marls and sapropel are mottled and gradational, from dark grey to dark green to dark brown.

Sampling was done in two successive field trips. During the first trip, we sampled a stratigraphic interval of 80 cm in which the reversal had been located earlier [*Zachariasse et al.*, 1989] and which consisted of a lower white marl, a grey marl including a sapropelitic layer, and an upper white marl. During the second trip, an interval of 25 cm where we had pinpointed the actual reversal was resampled to obtain more detail. Only a short interval of less than 5 cm within the sapropel could not be sampled due to fragility (lamination) of the sediment. We have sampled by taking oriented cores of 25 mm diameter, more or less parallel to the bedding plane at very close intervals (<1 cm) and from a freshly cut vertical plane. In order to get a higher stratigraphic resolution a number of specimens were split into upper and lower halves. The stratigraphic position of each specimen is accurately determined by taking into account drilling orientation, bedding plane and width of the saw cut. The stratigraphic levels in the present paper are in centimeters with respect to the bottom of the sapropel (Figure 2). The sampled upper Kaena interval therefore ranges from 25.07 to 25.87 m in Figure 2.

ROCK MAGNETISM

The upper Kaena reversal occurs in a layered lithology. In particular, this reversal is pinpointed close to a sapropelitic layer which has a lithology distinctly different from that of the surrounding marls. It is therefore important to ensure that the carriers of the characteristic remanent magnetization are of primary origin even though they may not be exactly the same in the various lithologies. Rock magnetic parameters give an indication of the content and nature of the magnetic minerals, and for this reason we have determined the initial susceptibility (X_0) and several (remanent) hysteresis parameters: the (remanent) saturation magnetization (J_{rs}, J_s) and (remanent) coercive force (H_{cr}, H_c).

Susceptibility and Hysteresis Parameters

Acquisition of isothermal remanent magnetization (IRM) of a number of samples from different lithologies up to a maximum DC field of 2 T (Figures 3*a* and 3*b*) reveals that samples outside the sapropel in general show a saturation IRM (J_{rs}) in fields of 125-200 mT. Samples from the sapropel itself are saturated at higher fields (0.5 T) and, in addition, have higher J_{rs} values (Figures 3 and 4). The sapropel samples appear to have both a low- and a high-coercivity IRM acquisition curve. The low-coercivity component has a saturation field similar to that of the nonsapropel samples. The relative intensity of this component may be estimated at approximately 20-30% of the intensity at 0.5 T if it is assumed that the J_{rs} acquisition of the high-coercivity component is linear in low fields [*Dekkers*, 1988] (Figure 3). This results in an absolute J_{rs} intensity of the low-coercivity component which is of the same order as the J_{rs} of the samples with only one component. Thermal demagnetization of the J_{rs} shows that approximately 95% of the total saturation remanence

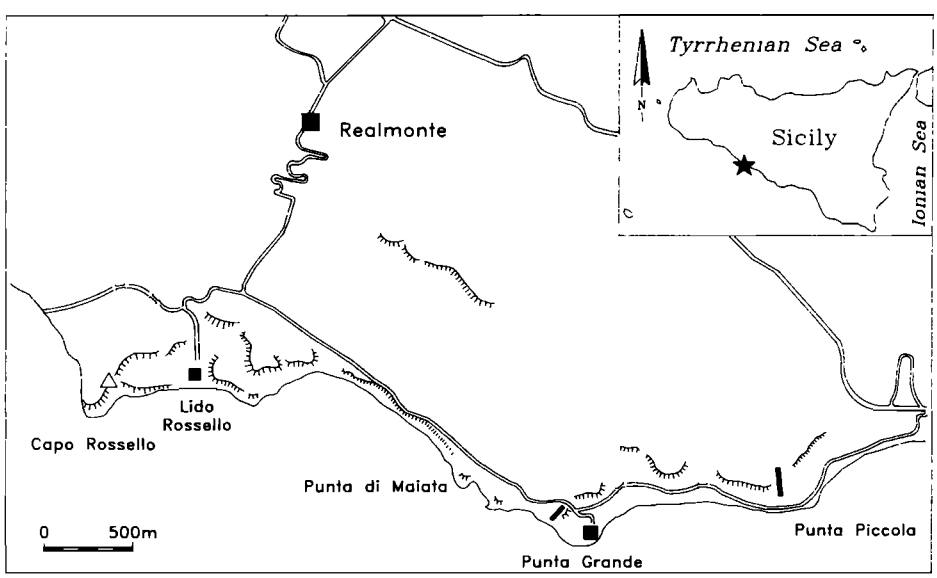


Fig. 1. Location of the Punta Piccola section in Sicily (Italy) in which the upper Kaena interval was sampled.

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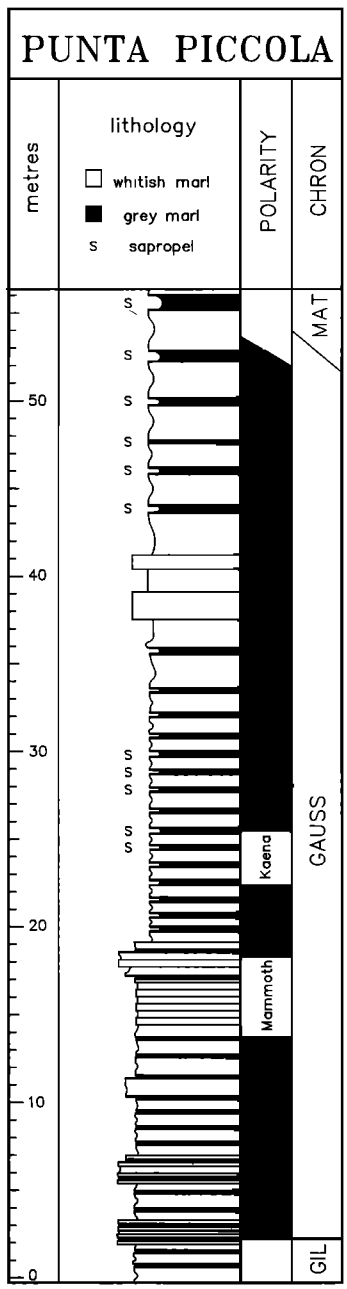


Fig. 2. Polarity sequence and lithology of the Trubi and Monte Narbone formations in the Punta Piccola section [after *Hilgen*, 1987; *Zachariasse et al.*, 1990] The 19-m level corresponds to the Trubi-Narbone transition. The upper Kaena reversal (indicated by an arrow) was sampled in detail over an interval of 80 cm, from 25.07 to 25.87 m. In the lithological column, the white layers are internally composed of a white, beige, and a white marl, the black intervals corresponds to a grey marl. The white marls have a higher carbonate content, while the grey and beige marls are relatively poor in carbonate [*Hilgen and Langereis*, 1989], probably due to increased continental runoff (grey) or increased African wind-blown input (beige) [*de Visser et al.*, 1989].

unblocks up to 580°C (Figure 3*c*). Together with the relatively low saturation fields (125-200 mT), this points to a magnetite dominated magnetic mineralogy. The remaining 5% shows maximum blocking temperatures of 580°-600°C, slightly higher than the maximum blocking temperature for pure magnetite. This suggests the presence of some cation-deficient magnetite [*Heider and Dunlop*, 1987; *Linssen*, 1988, 1991].

J_{rs} shows a clear lithological dependence: in the white layers, both above and below the grey layer, values are constant (3.5 A/m), while in the grey layer there is an increase (to 8 A/m) culminating in a maximum (of 15 A/m) in the middle of the sapropel. The lithological dependence of the initial susceptibility X_0 (Figure 4) is similar to that of J_{rs}: values are constant (250 x 10^{-6} S.I.) in the white layers and show a maximum (500-600 x 10^{-6} S.I.) in the sapropel. Neither X_0 nor J_{rs} is a useful magnetomineralogical indicator on its own because each is dependent on the nature and concentration of the magnetic minerals. The ratio J_{rs}/X_0 is more useful: it is largely independent of concentration, and provided that the dominant magnetic mineral is magnetite, it may give an indication of the grain size. The ratio shows typical values of 10-15 kA/m (Figure 4), lower than expected for fine-grained magnetite (>20 kA/m) [Dunlop, 1986]. In the grey layer and especially in the sapropel, values are significantly higher (15-25 kA/m).

However, the clay fraction has a relatively large influence on the bulk susceptibility X_0 because of its paramagnetic contribution. Hysteresis loop experiments show that this contribution is approximately 110 x 10^{-6} SI in the white layers, and as much as 250 x 10^{-6} SI in the sapropel (see Figure 5). Correction of X_0 for the susceptibility of the clay fraction does not significantly change the observed trend in X_0, but it causes the values for J_{rs}/X_0 to increase to 20-25 kA/m in the white layers and to as much as 50 kA/m in the sapropel. This clearly points to fine-grained magnetite as the dominant magnetic mineral in the white layers, whereas the high values in the sapropel point either to even finer-grained magnetite or to an additional magnetic mineral.

The remanent coercive force H_{cr} is largely independent of the concentration of magnetic material, and it is not influenced by paramagnetic clay minerals. Typical values for fine-grained magnetite are 40-60 mT [*Day et al.*, 1977; *Hartstra*, 1982; *Dunlop*, 1986], which are the values observed in the white layers (40-50 mT). In the grey layer/sapropel, H_{cr} values increase to 75 or even 100 mT which again indicates either even finer-grained (SD) magnetite or an addition of a high coercivity mineral (e.g., hematite, goethite). Although pyrrhotite also has a higher H_{cr} than magnetite for the expected grain size range, the observed maximum values of 90-100 mT would imply that the magnetomineralogy in the sapropel would be entirely dominated by this mineral [*Dekkers*, 1988], which is not the case.

The ratio $H_{cr}/(J_{rs}/X_0)$ [*Thompson and Oldfield*, 1986] is diagnostic for many magnetic minerals and may give useful information on the carriers of the remanence. The observed ratio $H_{cr}/(J_{rs}/X_0)$ is somewhat high (2-3) for very fine grained magnetites (Figure 4). Correction for the paramagnetic susceptibility of the clay fraction yields values of 1.2 in the white layers, compatible with fine-grained magnetite. Corrected values for the sapropel are only slightly higher, approximately 1.5, and in combination with the observed higher H_{cr} values (up to 75-100 mT) possibly indicate a trace amount of goethite [*Dekkers*, 1988]. Addition of goethite causes a relatively strong increase of $H_{cr}/(J_{rs}/X_0)$, whereas addition of pyrrhotite would decrease this

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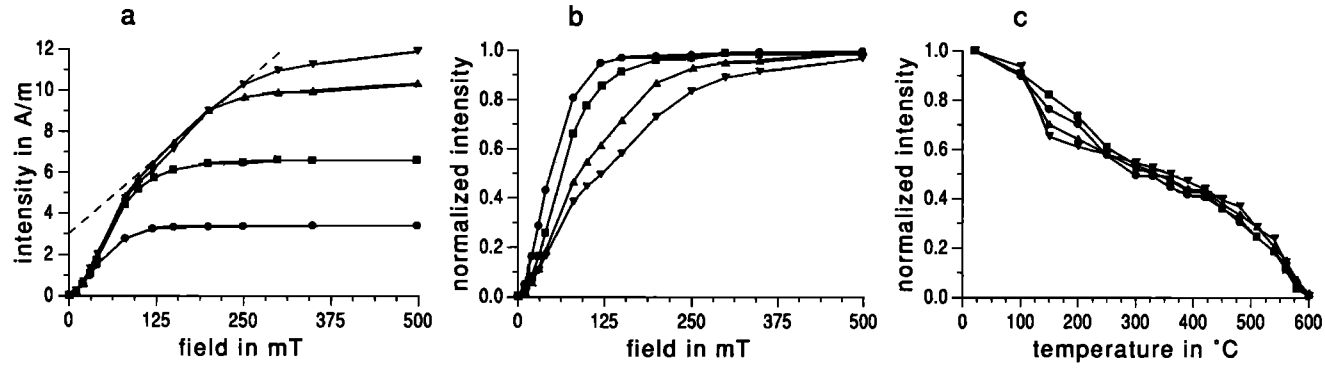


Fig. 3. (*a*) Absolute and (*b*) normalized IRM acquisition curves of samples from different lithologies, up to a maximum field of 1600 kA/m (2 T). Samples from the white marl (circles, level -22.8 cm) and from the grey marl (squares, level -5.1 cm) are saturated at 100-150 kA/m. Samples from the sapropel (triangles and inverted triangles, levels 1 and 2 cm, respectively) are only saturated at 400 kA/m. The curves show a low- and high coercive component. The low-coercive component is saturated ~100 kA/m; linear extrapolation of the high-coercive component back to the ordinate (dashed line) shows that the low-coercive contribution of the total IRM is 20-25%. (*c*) Subsequent thermal demagnetization of IRM shows similar decay curves except between 100 and 250 °C. Maximum unblocking temperatures are close to 580 °C, but a remaining 5% unblocks at slightly higher temperatures, possibly indicating some cation-deficient magnetite.

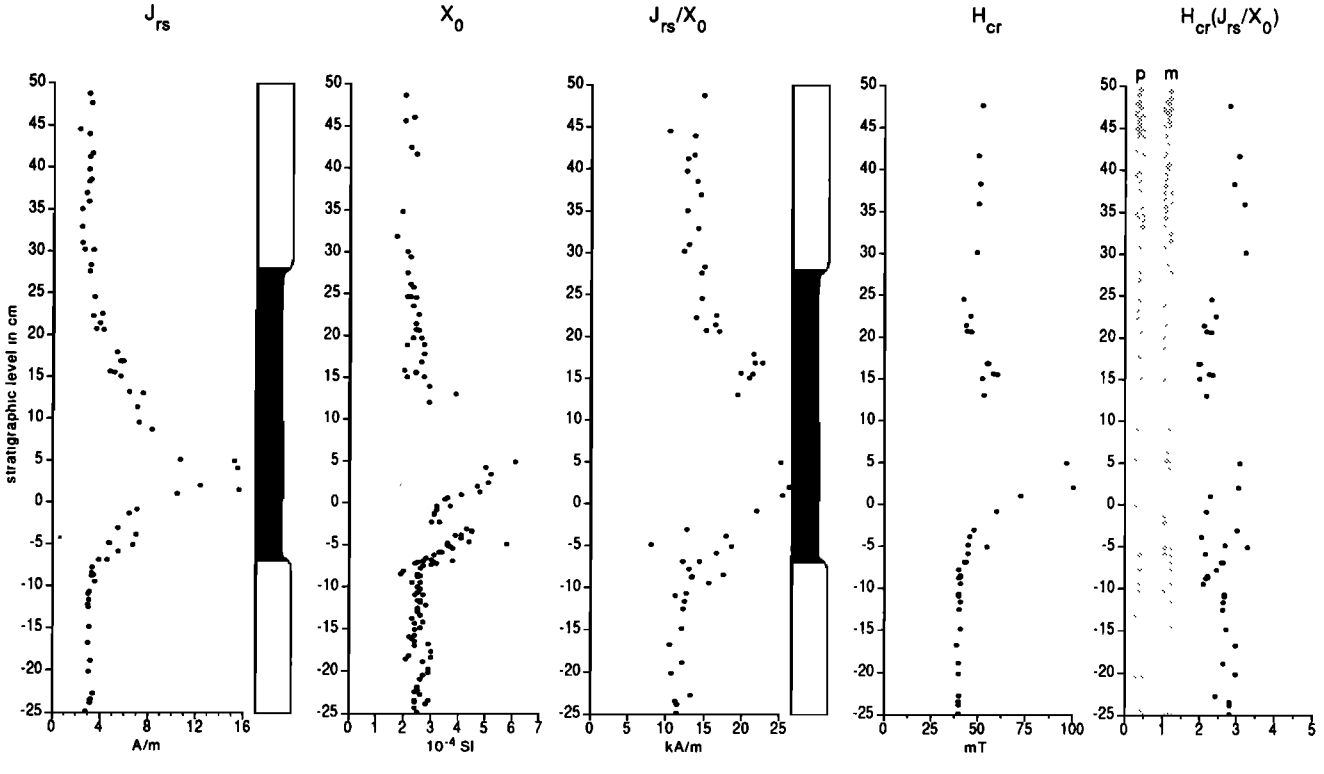


Fig. 4. Saturation IRM (J_{rs}), initial susceptibility (X_0), remanent coercive force (H_{cr}), and some interparametric ratios versus stratigraphic level and lithology: white marls (white), and a grey marl (black) including a sapropelitic layer (shaded). J_{rs} and X_0 show constant values in the white marls but reach a maximum in the grey/sapropelitic layer. Their ratio J_{rs} / X_0 is not constant either, indicating an addition of finer-grained magnetite and/or a (high-coercive) mineral. A similar trend is found for H_{cr}; the high maximum (100 mT) probably indicates addition of high-coercive material. This is supported by the $H_{cr}/(J_{rs}/X_0)$ ratio which shows a rather constant value (ranging 2-3) throughout the entire lithology. The ratio is dimensionless, since H_{cr} has been converted from mT to kA/m. The shaded lines denote typical values for fine-grained magnetites (m) and pyrrhotites (p).

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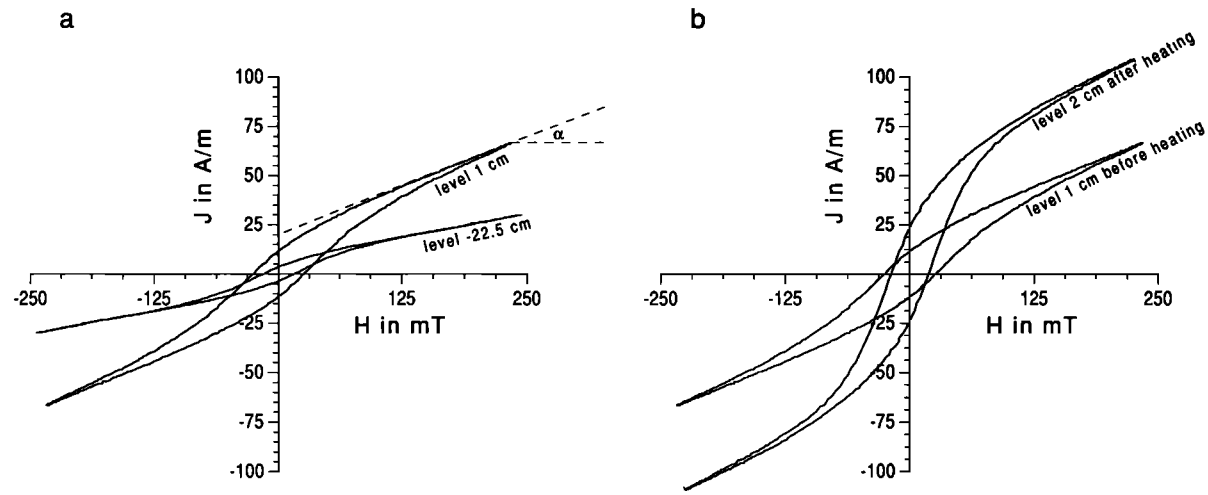


Fig. 5. (*a*) Hysteresis loops of samples from the white marl (-22.5 cm) and sapropelitic layer (1.0 cm). Saturation occurs at approximately 100 kA/m (-22.5 cm) and at 200 kA/m or higher (1 cm). The linear trend at higher fields (dashed lines) is due to the paramagnetic susceptibility of the clay minerals. It can be derived from these curves as the tangent (of α) and is approximately 110 (-23.9 cm) and 250 x 10^{-6} S.I. (1.0 cm). (*b*) Hysteresis loops of sapropel samples before heating (1.0 cm, same sample as in Figure 5*a*) and after heating to 360°C (2.0 cm). After heating, the sapropel samples have a higher saturation remanence.

ratio. This makes it rather unlikely that pyrrhotite is the mineral responsible for the observed changes in X_0, J_{rs}, and H_{cr}, despite the relatively high sulphur content in a sapropel [*Calvert*, 1983]. The presence of goethite, on the other hand, need not be surprising: it may be due to increased weathering in the dark brown and more permeable (laminated) sapropel. A similar line of reasoning with respect to the $H_{cr}/(J_{rs}/X_0)$ ratio excludes a possible small addition of hematite [*Dekkers and Linssen*, 1989].

Hysteresis loops (Figure 5*a*) show that a nonsapropel sample is saturated in a field of approximately 125 mT, whereas a sapropel sample is hardly saturated in the maximum field of 250 mT. This supports the idea that the sapropel samples may contain some high-coercivity mineral, i.e.,goethite. since these samples are not entirely saturated, only minimum values of J_S and H_C can be determined; complete saturation would slightly increase these estimates, corresponding with a decrease of the ratios H_{cr}/H_C and J_{rs}/J_S of the sapropel samples. The ratios H_{cr}/H_C and J_{rs}/J_S (table 1) of the nonsapropel samples are similar to the ratio's found by *Van Velzen and Zijderveld* [1990] who explain the discrepancy between their data and those of *Dunlop* [1986] by the presence of goethite and superparamagnetic (SP) magnetite grains. From the sapropel samples only, the ratios of the level 1.0 cm sample fit the data from *Van Velzen and Zijderveld* [1990], whereas the ratios from level 4.9 cm do not.

So far the conclusion that can be drawn from the observed rock magnetic parameters and some interparametric ratios is the likely presence of fine-grained magnetite in the white layers and even finer-grained magnetite in the grey/sapropelitic layer together with some (high-coercivity) magnetic mineral, possibly goethite. Without some additional data, for example, from X ray diffraction or microprobe analysis of magnetic concentrates, these rock magnetic parameters are not sufficient to make a more exact discrimination of the magnetic mineralogy. It is clear, however, that the grey layer and especially the intercalated sapropel show properties that markedly deviate from the underlying and overlying part of the record.

Susceptibility and Hysteresis Parameters During Thermal Demagnetization

During thermal demagnetization of the natural remanent magnetization (NRM), a number of specimens showed some scatter at temperatures higher than approximately 360°C. A small number of samples were heated to this temperature, and subsequently the rock magnetic measurements described above were repeated (Table 1). Most conspicuous is that after heating to 360°C also the sapropel specimens are also saturated at 125 mT, a value which is equal to the saturation fields of the nonsapropel samples before heating. This may be due to the removal of existing goethite.

J_{rs} itself does hardly change outside the sapropel, whereas in the sapropel, one sample showed a minor decrease and another sample showed a very strong increase (Table 1). The susceptibility after heating ($X360$) increases in the entire section with respect to X_0. The relative decrease of about 25-30% of $J_{rs}/X360$ with respect to J_{rs}/X_0 is the same for sapropel and nonsapropel samples. Growth of magnetite grains may have taken place in all specimens.

H_{cr} values from the sapropel, which showed maxima before heating, now show almost the same values as the nonsapropel samples. At the same time, all post-heating H_{cr} values are lower than those before heating. This is probably due to the removal of goethite and/or growth of magnetite grains: H_{cr} values decrease only slightly in the white layer but appreciably in the grey layer/sapropel.

In general, most rock magnetic properties that were characteristic for sapropel samples before heating, disappear

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TABLE 1. Rock Magnetic Parameters From the Upper Kaena Interval, Before and After Heating to 360°C

Level, cm	Saturation, kA/m		J_{rs}, A/m		X_0, 10^{-6} S.I.		J_{rs}/X_0, kA/m		H_{cr}, kA/m		$H_{cr}/(X_0/J_{rs})$		H_{cr}/H_c		J_{rs}/J_s	
	Before	After	Before	After	Before	After	Before	After	Before	After	Before	After	Before	After	Before	After
41.6	200	100	3.2	2.7	240	250	13.3	10.8	40	26	3.03	2.44				
24.5	100	100	3.4	3.5	240	300	14.1	11.6	33	29	2.33	2.50				
20.7	100		3.6		280		13.0		35		2.70					
13.0	100	100	7.4	7.3	390	420	18.9	17.2	42	29	2.22	1.69				
4.9	400	100	16.9	14.1	610	670	27.8	20.8	78		2.78		4.19	2.00	0.48	0.56
2.0	400	100	12.3	20.4	470	1000	26.3	20.4	80	30	3.03	1.47		2.00		0.41
1.0	400		10.4		410		25.6		58		2.27		2.90		0.42	
-5.1	150	100	6.7	7.0	360	650	18.5	10.9	44	30	2.38	2.78		2.54		0.26
-13.0	100	100	3.0		250		12.2		32		2.63					
-23.0	100		3.4	3.4	260	370	13.2	9.3	32	30	2.44	3.23		2.61		0.38
-24.0	100		3.2		280		11.4		32		2.78		2.56		0.40	

Area from 4.9 to 1.0 cm is the sapropelitic layer. The ratio $H_{cr}/(X_0/J_{rs})$ is dimensionless, since H_{cr} has been converted from mT to kA/m. See text for further discussion.

upon heating to 360°C. The resulting changes in magnetic parameters are mostly due to growth of (preexisting) magnetite grains during heating. In the sapropel some goethite may be present. The presence of pyrrhotite cannot be ruled out, but considering, for example, the $H_{cr}/(J_{rs}/X_0)$ ratio, it is thought to play a relatively minor role. Although the rock magnetic parameters from the grey/sapropelitic layer are somewhat atypical with respect to the parameters of the white layers, the dominant carrier of the remanence is magnetite. This conclusion agrees with the results of thermal demagnetization (see next section). *Van Velzen and Zijderveld* [1990], in a rock magnetic study of the very similar Trubi marls from the Eraclea Minoa section in southern Sicily, also concluded that the most important magnetic mineral in these marls is fine-grained (SD) magnetite, with a small amount of goethite.

NRM COMPONENTS

Measurements of the natural remanent magnetization (NRM) were done with a three-axis vertical 2G Enterprises cryogenic magnetometer. Progressive stepwise thermal demagnetization was carried out using a μ-metal shielded tunnel furnace. In general, small temperature increments of 30° or 50°C were used. The demagnetization properties of the Pliocene marls from the Trubi and Monte Narbone formations are generally very constant, and we first give a general description before discussing the details of the upper Kaena interval.

NRM Components in the Trubi and Monte Narbone Marls

Typical thermal demagnetization diagrams of the Pliocene marine marls from southern Sicily generally show several distinctly different components [see *Langereis and Hilgen*, 1991]. Often, a small component is removed at low temperatures (90-120°C); since its direction is mostly related to storage, a viscous (laboratory induced) origin is likely. A component removed at 200°-250°C has a present-day field direction: it is easily distinguishable from the characteristic remanence removed at higher temperatures since it has a steeper inclination and a northerly declination, whereas the characteristic remanence shows a consistent 35° (average) rotation during the early Pliocene which reduces to 25° in the upper part of the Gauss Chron. A characteristic remanent magnetization (ChRM) is removed between 200°-250° and 580°-610°C. This ChRM actually consists of two components [*Langereis and Hilgen*, 1991], a low-temperature (LT) component removed between 200°-250° and 480°-510°C and a high-temperature (HT) component removed between 480°-510°C and the highest temperatures used. The LT component decays most strongly between 200° and 330°-360°C. Often, there is almost no decay between 360° and 480°C, but some scatter is frequently observed which is probably due to magnetic minerals being formed during heating. The origin of the LT component is uncertain, but the strong decay between 200° and 330°-360°C may indicate that either pyrrhotite or greigite is a carrier of the remanence in these marls [see *Leslie et al.*, 1990; *Linssen*, 1988, 1991]. The HT component decays most strongly between 540° and 580°C, resulting in a distinctive and narrow peak in the blocking temperature spectrum, which may be characteristic of ultra-fine-grained SD magnetite in a narrow grain size range and possibly of biogenic magnetite [cf. *Chang and Kirschvink*, 1985; *Lovley et al.*, 1987; *van Hoof and Langereis*, 1991].

Typically, outside polarity transition intervals, the LT and HT component have the same direction, including the characteristic tectonic rotation of the Caltanisetta basin. Inclinations of the ChRM are consistently shallower than the inclination of the present-day field at the locality. This inclination error is most likely due to sediment compaction [see *Laj et al.*, 1982; *Anson and Kodama*, 1987; *Arason and Levi*, 1990] since the marls with a higher carbonate content [cf. *Hilgen and Langereis*, 1989] and thus less compaction consistently show a smaller inclination error. This inclination error may be taken as an indication of an early origin of the magnetization [*Laj et al.*, 1982]. Together with the fact that the observed polarity pattern in these Pliocene marine marls shows an excellent correlation with the polarity time scale [*Zijderveld et al.*, 1986; *Langereis and Hilgen*, 1991],

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these observations provide strong evidence that the ChRM has been acquired in a very early stage after deposition.

NRM Components in the Upper Kaena Interval

Throughout the upper Kaena interval, the remanence removed between 200° and 330°-360°C is small, some 15-20% of the total ChRM intensity (Figure 6*a* and 6*f*), whereas usually values of 30-50% are found. since the maximum blocking temperatures of pyrrhotite are within this temperature range [*Schwarz*, 1968, 1975; *Dekkers*, 1988], this supports our conclusion that in the upper Kaena interval (almost) no pyrrhotite is present. In some samples, the remanence direction starts to fluctuate at temperatures higher than approximately 360°C up to 480°-510°C (cf. Figure 6*a*). These fluctuations are probably caused by growth of existing magnetite grains and/or by production of SP magnetite grains from clay minerals during laboratory heating. This is illustrated by the often large increase of the magnetic susceptibility during stepwise thermal demagnetization at temperatures of 390°-420°C (Figure 7). At temperatures higher than 450°-480°C susceptibilities decrease again but remain always higher than values at room temperature. Remarkedly, susceptibilities of samples from the sapropel do not show the clear maximum between 390° and 480°C, but gradually increase to values some 30% higher than those at room temperature (Figure 7).

If these fluctuations in NRM direction take place, the direction of the LT component is determined by least squares fitting [*Kirschvink*, 1980] of demagnetization steps between 200°-250°C and 360°-390°C, if not, from steps between 200°-250°C and 450°-480°C. The direction of the HT component is ,where possible, determined from demagnetization steps higher than 480°-510°C; the origin was always included in the least squares fitting, even if the HT component does not clearly converge towards the origin (as in Figure 6*d*). The strong remanence decay of samples outside the reversal between 510°-540°C and the maximum temperature (Figures 6*a*, 6*b* and 6*f*) indicates fine grained magnetite as a carrier of the HT component. Maximum blocking temperatures (590°-610°C) are higher than expected for magnetite. This is probably not due to the presence of hematite, as discussed in the section on rock magnetism (and see, e.g., Figure 3) but may rather be due to the presence of cation-deficient magnetite [*Heider and Dunlop*, 1987; *Linssen*, 1991]. The majority of the vector diagrams with intermediate directions

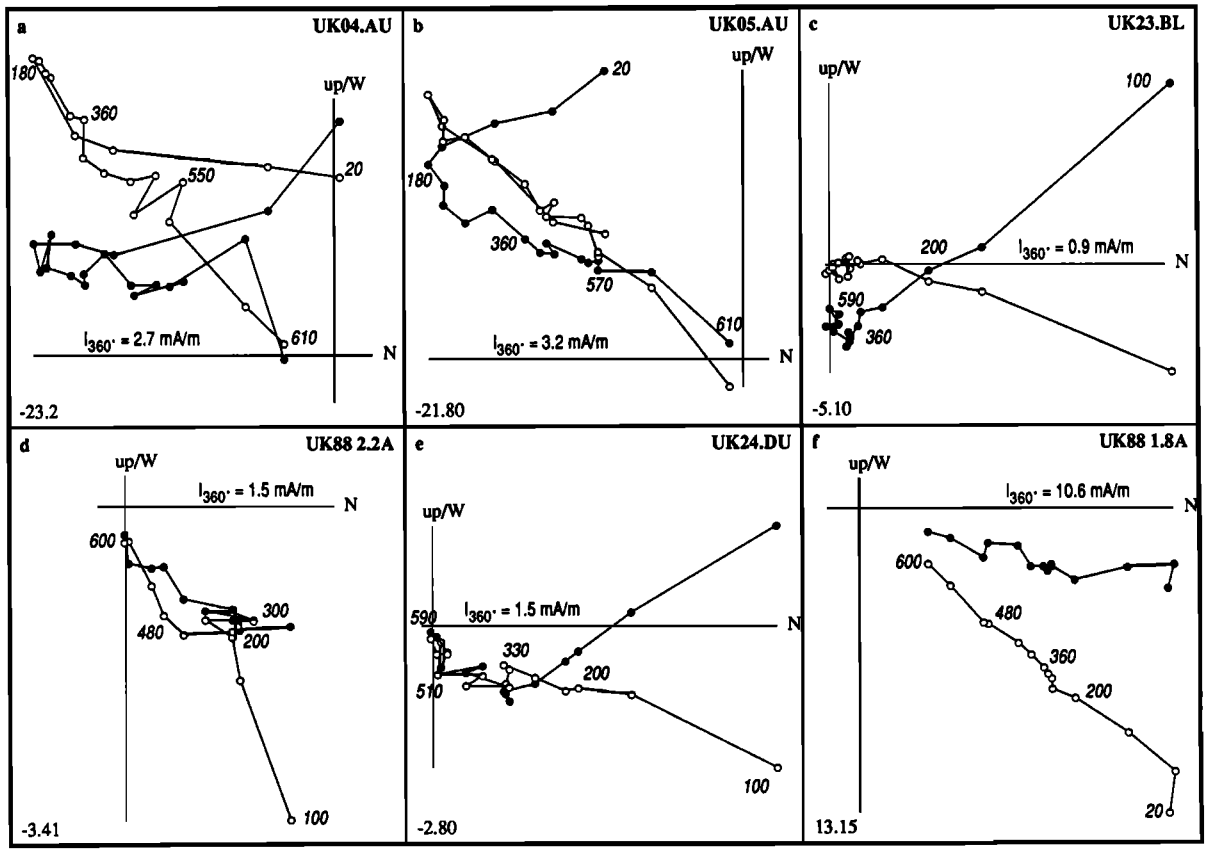


Fig. 6. Thermal demagnetization diagrams from the upper Kaena interval. The intensity at 360°C ($I_{360°C}$) and the stratigraphic level (in centimetres; down left) is given in each diagram; open (solid) symbols denote projection on the vertical (horizontal) plane; numbers are temperatures in degrees Celcius. (*a*) and (*b*) Stable reversed directions show some scatter at temperatures higher than 390°C due to growth of newly formed magnetite (cf. Figure 7) but (*f*) less noticeable in stable normal directions. (*c*), (*d*), and (*e*) Intermediate directions show a difference in low-temperature (LT, 200°-360°C) and high-temperature (HT, 480°-580/610°C) components.

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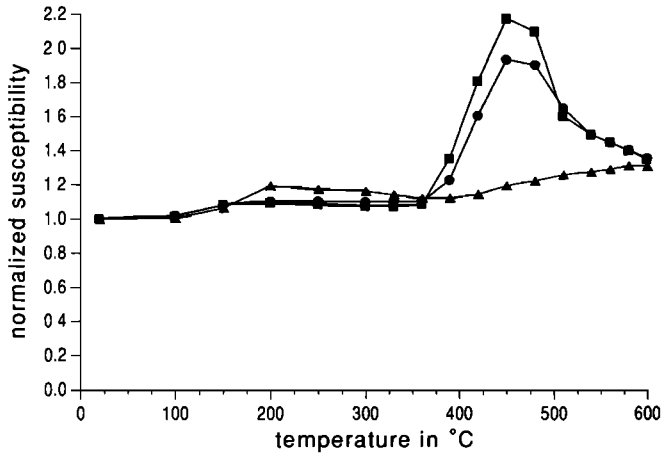


Fig. 7. Susceptibilities, normalized to their value at room temperature, during stepwise thermal demagnetization of nonsapropel samples (circles, squares) show a considerable increase between 360° and 450°C, and decrease again at higher temperatures. This probably indicates that growth of magnetite grains occurs, and/or that magnetite is formed from clay minerals during heating. A sapropel sample (triangles) does not show a maximum but more or less steadily increases to a 30% higher value.

(Figures 6*c*, 6*d* and 6*e*) are not as well behaved as the nontransitional diagrams, probably because the ChRM intensity is lowered during the reversal: a same amount of scatter due to growth of magnetite will then have a much stronger disturbing influence.

The nontransitional reversed (Figures 6*a* and 6*b*) and normal (Figure 6*f*) ChRM directions from the upper Kaena interval both show a distinct clockwise rotation. The mean stable, nontransitional directions from the upper Kaena interval (Figure 8) agree reasonably but not quite with the mean directions of the entire underlying and overlying polarity zones from the Punta Piccola section (Table 2). The actual angle between the mean reversed and normal stable directions of the upper Kaena interval is 11.5°, whereas this difference is only 4.6° in the Punta Piccola section.

The demagnetization diagrams within the transitional interval (Figures 6*c*-6*e*) show that LT and HT components seem not to have the same (ntermediate) direction. Therefore, we have determined, where possible, both the LT and HT components separately throughout the entire sampled interval (Figure 9). In nontransitional samples (Figures 6*a*,6*b*,6*f*) the secondary component is clearly different from the LT + HT component. It may be argued that in transitional samples (Figures 6*c*-6*e*) the LT component is more strongly influenced by the secondary component due to the relatively low LT intensities during a reversal, while the intensity of the secondary component remains approximately the same. However, *van Hoof and Langereis* [1991] showed that during two older (N-R) reversals (lower Cochiti and lower Nunivak) from the same marine marls the LT component is reversed, and thus not a recent overprint, while the HT component is normal, and concluded that there is a magnetization depth lag between the two components. The very similar magnetic properties between the just mentioned records and the present record as well as the clearly different directions of secondary and LT components in nontransitional samples give us no a priori reason to assume that the LT component represents a (partial) overprint by secondary (subrecent) remanences in transitional samples. Admittedly, the coincidence of a sapropel being present in the upper Kaena record may give rise to increased weathering, but it can be seen that just in this sapropel LT and HT components have the same characteristic direction (Figure 9).

Although very rapid changes have been reported both in sediments and in lava flows [*Laj et al*., 1988; *Coe and Prévot*, 1989], we feel that a single ChRM direction strongly deviating with respect to adjacent samples or an unreliable direction, due to low intensity and/or large scatter, must be interpreted with caution. Such directions are denoted by diamonds in Figure 9 and were rejected in the calculation of the VGP paths.

THE UPPER KAENA REVERSAL RECORD

Nontransitional Stable Directions

Outside the transitional interval, the directions of the HT component follow the same trend as the LT component. Mean directions of the HT component from the upper Kaena interval immediately before and after the reversal are not antipodal (by 11.5°), especially with respect to the declination (Table 2). They

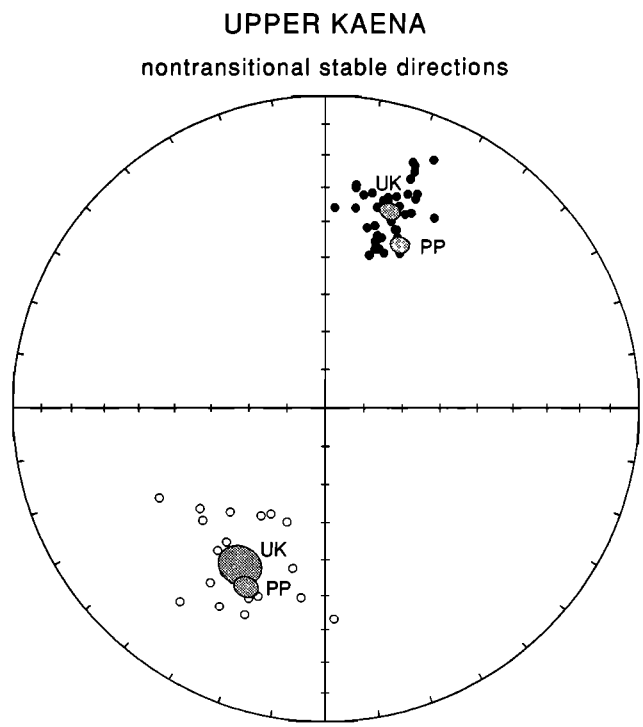


Fig. 8. Nontransitional stable reversed (open circles) and normal (solid circles) directions from the upper Kaena (UK) interval. The mean directions including their $\alpha 95$ are given by the shaded area. For comparison, the mean directions of the entire underlying and overlying polarity zones from the Punta Piccola (PP) section are also given. See also Table 2.

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TABLE 2. Mean Normal (N) and Reversed (R) Stable (Nontransitional) Directions From the Upper Kaena (UK) Interval and From the Entire Underlying and Overlying Polarity Zones From the Punta Piccola (PP)Section

	n	D, deg	I, deg	k	α_{95}	r	γ_0	γ_c	ctmd
UK (N)	45	18.1	34.2	80.8	2.4	44.46			
							11 5	5.9	-
UK (R)	23	208.6	-42.3	33.1	5.3	22 33			
PP (N)	32	24.5	42 2	122.3	2.3	31.75			
							4.6	4.1	-
PP (R)	12	203.9	-37.6	219.2	2.9	11.95			
UK-PP (N)							9.4	3 1	-
UK-PP (R)							5.9	6.1	B

In both cases, the reversal test, denoted as ctmd (common true mean direction [see *McFadden and Lowes,* 1981] is negative Also, the mean normal directions from the upper Kaena interval and the Punta Piccola section do not share a ctmd, while the mean reversed directions do, with classification B. Reversal (or ctmd) test and classification have been determined using simulation after *McFadden and McElhinny* [1990], except in the case of PP (N) versus PP (R) The actual (γ_0) and critical angle (γ_c) between every two tested populations are given

show a negative reversal test [*McFadden and McElhinny*, 1990] (Table 2). Average inclinations of the HT component (-42.3° and 34.2°) are shallower than the geocentric axial dipole field inclination for the present latitude (58°). The directions of the LT component show considerably more scatter, which may be mainly due to the lower intensity of this component (Figure 9) and the consequent difficulty in determining a reliable direction. We have tested whether the mean reversed (normal) stable direction from the upper Kaena interval and the mean reversed (normal) direction from the entire polarity zones in the Punta Piccola section have a common true mean direction [*McFadden and Lowes*, 1981] (Figure 8 and Table 2]. We have done this by inverting one of the two reversed (normal) directions and subsequently using the reversal test of *McFadden and McElhinny* [1990]. This test uses the same statistics as the earlier test of *McFadden and Lowes* [1981], except that it may be performed by simulation if the hypothesis of a common precision parameter must be rejected. The reversed directions have a common true mean direction, with classification B [*McFadden and McElhinny*, 1990], while the normal directions have not (Table 2).

Transitional Directions

The actual reversal is characterized by an eastward movement of the HT declination starting at level -15 cm; it changes via east (90°) at level -5 cm to normal polarities at 0 cm, where they show the familiar clockwise rotation. The HT inclination shows a gradual change from reversed to normal values; no steep inclinations are observed during the reversal. At level 0 cm the inclination is close to normal values. Inclinations of the LT component follow a similar gradual trend as HT inclinations, but

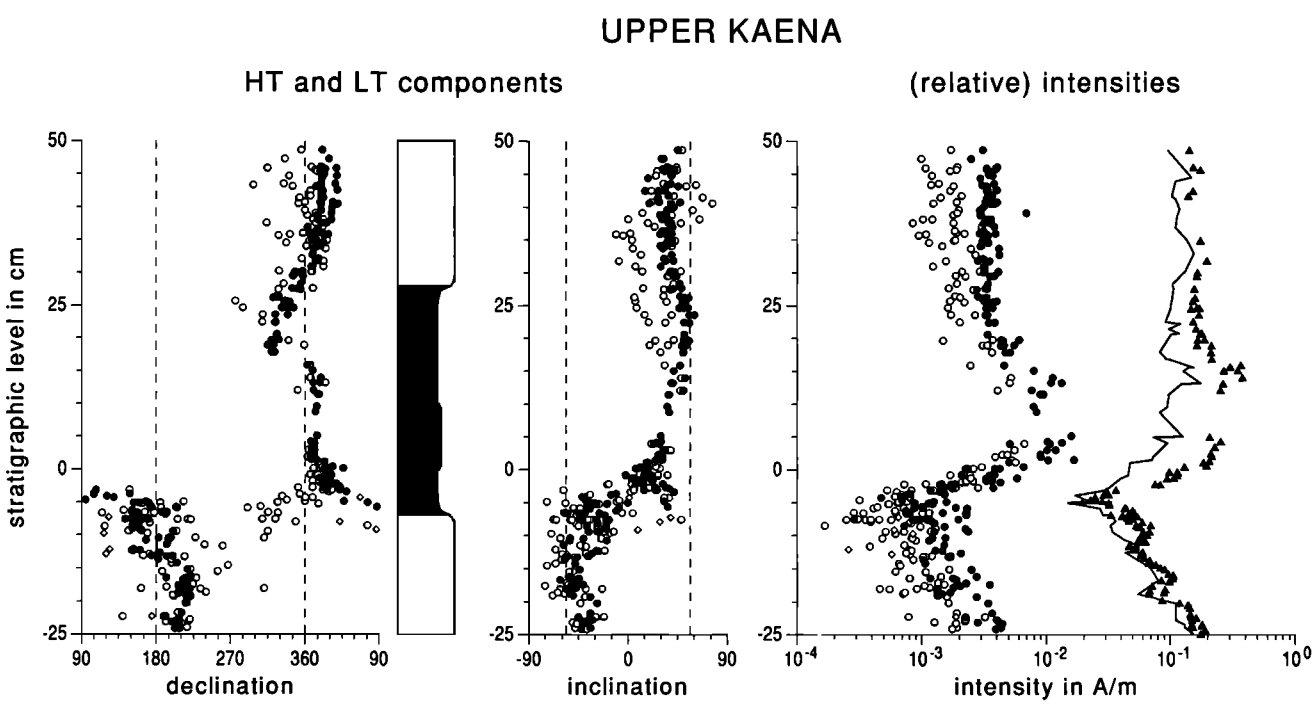


Fig. 9. Declinations, inclinations and intensities of the HT component (solid circles) and the LT component (open circles) during the upper Kaena transition interval. Diamonds denote unreliable directions (see text) and are not used for calculation of VGPs (Figure 10). Transitional directions, between -15 and 0 cm, proceed anticlockwise via east (HT component) or oscillate between east and west (LT component). After completion of the transition, an excursion in declination is seen between 17 and 30 cm; inclinations are slightly steeper. Relative intensities are denoted by triangles ($I_{360°C}$ / X_0) and by a solid line ($I_{360°C}$ / J_{rs}) and reach a minimum at the level of greatest directional changes, at -5 cm. During the excursion, no apparent changes in intensities are found.

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LT declinations differ from those of the HT component. Whereas the HT declination has a clear easterly component, the LT declinations show both easterly and westerly deviations during the transition. We note, however, that in this interval LT intensities are extremely low (Figure 9) and the reliability of the LT directions may be strongly reduced. The ChRM intensity starts to decrease at the -20 cm level, slightly before directional changes start to take place (at -15 cm). The intensity minimum is some 10% of prereversal and postreversal values and occurs at -5 cm (Figure 9), at exactly the same level of the largest directional changes. The intensity seems to regenerate to prereversal values at approximately 5 cm, slightly after directions have returned to normal values (at 0 cm).

The directions remain normal between levels 0 and 14 cm, and the average declination agrees well with the generally observed rotation. In this interval, the ChRM intensities show their maximum values.. Between levels 14 and 30 cm, the declination shows a considerable deviation of some 50°-60° to the west, and the inclination shows values which are generally some 15° steeper than the average inclination. It would seem that a directional excursion is recorded after the actual upper Kaena reversal. The ChRM intensities, however, do not show a noticeable decrease. Above 30 cm, the directions show a stable normal polarity direction which has a mean declination still less than expected for these marls in the Punta Piccola section (Table 2).

Relative Intensities

The observed changes in magnetic mineralogy over the reversal record do not allow the intensities of the LT and/or HT component to be used as a measure of the relative paleo-intensity of the geomagnetic field during the reversal. Usually, ARM is used as a normalizing factor to determine relative paleontensities (NRM/ARM ratio) [see *Levi and Banerjee*, 1976; *King et al.*, 1983]. On the base of earlier results of older reversal records in the same marine marls from Calabria and Sicily, the presence of pyrrhotite was suspected, in which case the method developed by *King et al.* [1983] cannot be used [*Linssen*, 1988, 1991]. Therefore, we routinely determined X_0 and J_{rs} (see Figure 4) rather than ARM for a large number of samples.

We have calculated the ratio of ChRM intensity (at 360°C) and X_0 as a first order approximation of the paleointensity (triangles in Figure 9). The magnetomineralogical effect of the sapropel seems not to be entirely eliminated since there is still a maximum corresponding to the maximum of the ChRM intensity (Figure 9) and other rock magnetic parameters (Figure 4). This is probably due to the fact that X_0 is largely determined by the paramagnetic clay fraction. Therefore, this ChRM/X_0 ratio is not representative for geomagnetic field intensity changes. since we have determined the paramagnetic contribution only for a few samples, we cannot use the (corrected) X_0 to normalize the ChRM intensity. Instead, we have used the ratio ChRM intensity (at 360°C) and J_{rs} intensity (at 2 T) as a measure of the relative paleointensity (line in Figure 9). The J_{rs} value (Figure 4) of the same or the closest possible stratigraphic level was taken to calculate this ratio for each specimen. The increase of ChRM intensity in the sapropel is now largely compensated by the increase of J_{rs} in the sapropel and it confirms that this increase of the ChRM intensity is a purely mineralogical effect.

Virtual Geomagnetic Poles

The virtual geomagnetic poles (VGPs) of both LT and HT components were calculated after applying an average correction of 25° for the clockwise rotation of the location, i.e., that of the Punta Piccola section. Correction for the inclination error was not applied. The HT component VGP path (Figure 10) initially follows roughly a path some 60° east of the site, loitering southwest of Australia, then quickly passes the equator etween India and Australia and goes via Japan to high northern latitudes approximately 180° east of the site. Furthermore, the VGP path shows an easterly migration of more than 120° during the reversal. The excursion following the reversal shows VGPs more or less antipodal to the VGP path of the actual reversal itself. They are confined to North America and remain at latitudes higher than 30°N. The LT component VGP path (Figure 10) shows a more complicated picture. Rather than following a particular great circle, it oscillates between two meridians: one passing over Australia and one passing over South and North America. Surprisingly, no VGPs intermediate between these two great circles are found. Initially, most VGPs are on the southern hemisphere of the Australian path while relatively few are found on the (South) America path. During the final stage, VGPs are only found on the (North) America path, while there are no VGPs on the northern hemisphere of the Australia path.

DISCUSSION

The upper Kaena record is the last reversal from the Punta Piccola section in a series of four successive reversals which were sampled by our laboratory. Recently, *Linssen* [1991] reported the records of the lower Mammoth, upper Mammoth and lower Kaena reversals. Such a sequence of successive reversals may yield useful information on the time dependent character of transitional field behavior during longer time spans, provided that the records reflect true geomagnetic field behavior.

The average sedimentation rate during the Kaena subchronozone is 4.3 cm/kyr on the basis of the magnetostratigraphic results [*Zachariasse et al.*, 1990] and their correlation to the polarity time scale *Berggren et al.*, [1985]. A more precise estimate is based on the exact location of both lower and upper Kaena reversal [*Linssen*, 1991; this paper] and the correlation of the individual sedimentary cycles to the astronomical (precessional) solutions [*Hilgen*, 1991], yielding an average sedimentation rate of 4.5 cm/kyr. The sedimentation rate during the deposition of the grey marls, including the intermittent occurrence of a sapropelitic layer, is probably slightly higher than the average 4.5 cm/kyr due to increased continental runoff [*de Visser et al.*, 1989]. Although slight changes in sedimentation rate will not essentially alter any duration of features in the upper Kaena reversal, they may cause noticeable changes in paleoredox conditions, as will be discussed later.

A conventional interpretation of the record would suggest that the major directional changes take place in a sedimentary interval of approximately 15 cm or 3.3 kyr, followed by an interval of stable normal directions (15 cm or 3.3 kyr) and the excursion (16 cm or 3.6 kyr). The period of lowered intensities associated with the reversal takes some 25 cm (5.6 kyr), while

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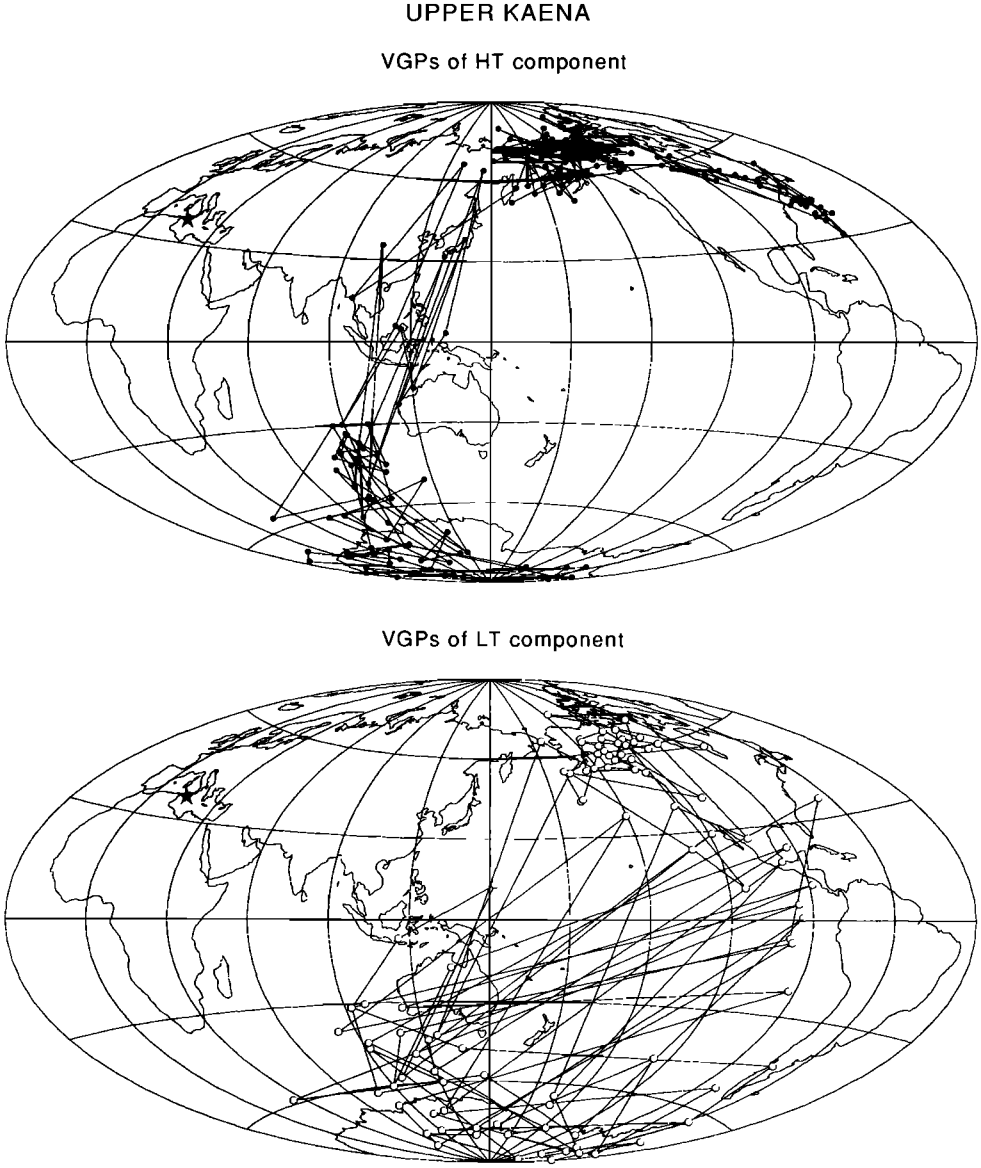


Fig. 10. Aitoff projection of virtual geomagnetic poles (VGP) of the HT and LT component from the upper Kaena reversal. Star denotes site location. The VGP path of the HT component shows a significant eastward migration during the reversal. The HT component VGPs of the "excursion" are restricted to eastern North America. The VGPs of the LT component oscillate between a path over India/Australia and a path over the Americas. VGPs of the excursion are similar to those of the HT component VGPs (not shown for readability).

the total interval from consistently stable reversed to stable normal directions takes 50 cm or 11 kyr. These durations seem to agree well with existing estimates of 4-10 kyr [e.g., *Clement and Kent*, 1984). However, before interpreting the reversal record in terms of geomagnetic field behavior, it is important to first establish the extent to which the present sedimentary record actually represents geomagnetic field changes.

Smoothing of the NRM

A major drawback of sedimentary reversal records is the lack of detailed knowledge of the NRM acquisition mechanism, although very recently a number of studies on important aspects related to the acquisition and recording process have been published [e.g., *deMenocal et al.*, 1990; *Lund and Karlin*, 1990; *Karlin*, 1990*a,b*; *Leslie et al.*, 1990; *Rochette*, 1990; *van Hoof and Langereis*, 1991].

A common mechanism by which many types of sediment, typically deep-sea sediments and lake sediments, acquire their remanence is postdepositional remanent magnetization (PDRM) [*Irving*, 1957], which is acquired below the bioturbated zone [0-10 cm) during early dewatering and consolidation of the sediment. Estimates for the PDRM lock-in zone range from 10 to 100 cm below the sediment surface. Recent data from the Brunhes-Matuyama reversal indicate a 42-cm magnetization depth lag (half fixing depth) with a lock-in zone of 10-20 cm in high deposition rate sediments (180-370 cm/kyr) [*Okada and*

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Niitsuma, 1989], and a PDRM acquisition depth of ~16 cm for eight moderately slow deposition rate deep-sea cores (1-8 cm/kyr) [*deMenocal et al.*, 1990]. Although PDRM acquisition inevitably involves smoothing of the geomagnetic record, it is often assumed that PDRM acquisition preserves the true geomagnetic record (albeit with a depth/time lag), provided sedimentation rates are high enough, at least more than 1 cm/kyr.

The smoothing process of reversal records has been discussed by various authors [e.g., *Hoffman and Slade*, 1986; *Rochette*, 1990]. The most evident type of smoothing is due to specimen thickness. Secular variation is generally assumed to be averaged out in moderately slowly deposited sediments; here, a single specimen represents some 500 years. However, the British secular variation master curve during the past 3000 years [*Clark et al.*, 1988] shows differences in declination of some 70° within this time span and occasionally differences of more than 40° within 500 years. If smoothing would occur due to sample thickness only, one would expect the secular variation to be expressed both before and after the reversal. In order to estimate the maximum amplitudes of secular variations, we have applied a 500-year filter to the British master curve, using time steps of 100 years, i.e., approximately the sampling resolution of the present record. It can be seen that amplitudes of more than 30° may then still be expected, although high frequency details are lost (Figure 11). Comparison with the nontransitional upper Kaena directions shows that prereversal stable directions show on average similar amplitudes, whereas postreversal stable directions show a maximum amplitude of 15° (Figure 11). However, it must be realized that the apparently similar amplitude of the prereversal directions is for a large part due to a few outlying data points. Although the large amplitudes present in the British master curve during the last 3000 years need not be representative for older time spans, we suspect that some additional smoothing, other than from specimen size only, has taken place. This additional smoothing is most probably related to the mechanism of remanence acquisition in these marine marls.

Smoothing will (strongly) affect short-scale features like secular variation, but the general pattern of the reversal may be preserved if the smoothing window is smaller than the reversal duration. Furthermore, it is often assumed that the smoothing window must be smaller than the observed duration of rapid directional changes. If this were true, we would arrive at a smoothing window of less than ~5 cm or 1 kyr, while the onset of the excursion even shows a 40° change in declination within 2 cm or 400 years (see Figure 9). However, *Hoffman and Slade* [1986] have shown that due to smoothing directional changes may appear more rapidly than in the original reversal record. Also *Rochette* [1990] has argued that in the case of nonantipodal stable directions and a smoothing window large compared to reversal duration, this assumption is not true. In such a case, the intensity change is observed over the width of the smoothing window, while the directional change, although smooth, is much more rapid: transitional directions follow a great circle defined by the nonantipodal stable directions. Nonantipodality mainly in inclination leads to a zonally confined path, while nonantipodal declinations produce a VGP path some 90° away from the site.

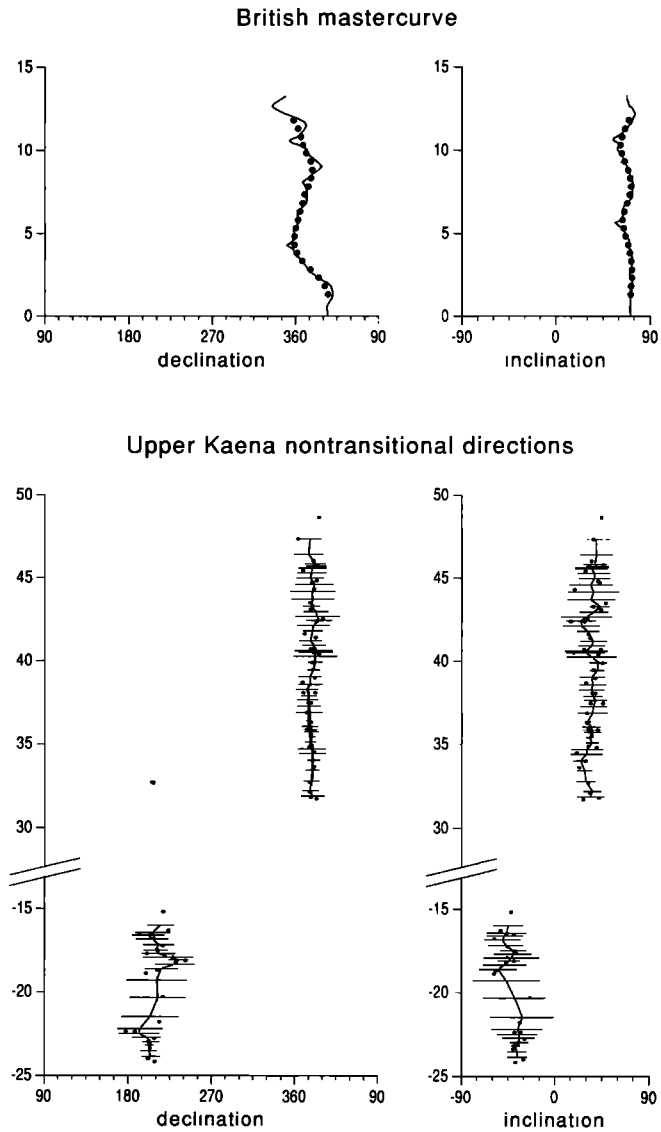


Fig. 11. British secular variation master curve (thin line) [after *Clark et al.*, 1988], sampled with a 500-year filter (dots) to represent specimen thickness, at time intervals of 100 years (sampling resolution). The time scale (1000 B.C. to 1950 A.D.) has been converted to stratigaphic level using a sedimentation rate of 4.5 cm/kyr. For comparison, also the stable pretransitional and posttransitional upper Kaena directions are plotted (dots). Line denotes a three-point moving average; bars indicate the three-point α_{95}. It can be seen that the directional variations of the upper Kaena interval are significantly less than those derived from the British master curve, indicating smoothing in addition to that resulting from specimen thickness only.

In our record, we find nonantipodal directions in the upper Kaena interval, mainly due to a ~10° difference in declination [Table 2). Indeed, intensity changes take longer than directional changes [5.6 and 3.3 kyr, respectively) and VGP paths follow a longitude well away from the site longitude (Figure 10). However, the HT component VGPs are located east rather than west of the site as should be expected, while LT component

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VGPs are found at both 90° east and 90° west. On the other hand, true nonzonal behavior is quite indifferent to filtering, and a commonly found feature of reversal records like a typically nonzonal and longitudinally confined path may indicate a relatively stable feature of the transitional field since both smoothing and true record give similar results [*Rochette*, 1990].

Many reversal records show the occurrence of loitering (VGP "hangups") and rapid rebounds/excursions which are not easily explained by smoothing, although slight variations in the shape of the averaging filter may account for these features [*Rochette*, 1990]. It is therefore important to discuss the acquisition process and corresponding lock-in mechanism in some more detail, since the rythmically bedded Pliocene marls may induce a complex sedimentary filter indeed.

NRM Acquisition Processes in a Cyclic Environment

Geochemical and rock magnetic investigations on whole sediment and magnetic concentrates from both low and high carbonate marls from the Calabrian Trubi formation by *Linssen* [1991] have shown that the major magnetic minerals (pyrrhotite, magnetite) in these marls must have been formed by authigenic processes in the early stages after sediment deposition. *van Hoof and Langereis* [1991] have suggested that these minerals are possibly of bacterial origin, because indications for biogenic magnetites were also found by *Chang and Kirschvink* [1985] in very similar upper Miocene marine marls from Crete [*Langereis*, 1984]. *Linssen* [1991] concluded that pyrrhotite, in coexistence with pyrite, is relatively more abundant in the low carbonate (grey) marls, while magnetite is relatively more abundant and less fine-grained (PSD rather than SD) in the high carbonate (white) marls. This may indicate that in the grey marls the progressive series of redox reactions involved in the decomposition of organic matter [*Froelich et al.*, 1979] has progressed into sulfate reduction, while in the white marls they are mainly restricted to Mn and Fe reduction. The cyclic variation in color, carbonate content, and rock magnetic properties strongly suggests that these lower Pliocene marine marls have been subject to fluctuations in [paleo)redox conditions. In the cyclic Monte Narbone marls the same processes have most probably played a role, considering the very similar sedimentary properties [*Hilgen*, 1987; *de Visser et al.*, 1989] and rock magnetic properties [*Linssen*, 1991; *Langereis and Hilgen*, 1991].

Both magnetotactic and dissimilatory iron reducing bacteria have been proposed as the main producers of ultra fine authigenic magnetite in anoxic sediments [*Lovley et al.*, 1987; *Bazylinski et al.*, 1988] as well as in suboxic sediments, at the top of the iron reduction zone [*Karlin et al.*, 1987; *Karlin*, 1990*a*]. In suboxic sediments deposited at moderate rates (1-5 cm/kyr) the reaction sequence does not seem to progress into sulfate reduction [*Karlin et al.*, 1987], although *Linssen* [1991] has observed both pyrite and pyrrhotite in magnetic concentrates from the Trubi marls. The finely laminated structure of the (top part of the) sapropel is due to the absence of burrowing and indicates oxygen depleted conditions. Suboxic conditions are furthermore in agreement with a possible higher organic input due to increased continental runoff [*de Visser et al.*, 1989]. Indeed, the sedimentological and rock magnetic properties of the grey and white layers show a striking similarity with those of suboxic sediments from the Bettis area in the NE Pacific which were studied in detail by *Karlin* [1990*a*]. The fresh colors of the grey marls and sapropel are the same (from dark/brown to green and grey) and, if representative of their original, primary colors during deposition, they indicate a preserved paleoshift in redox conditions from surface oxidation to Mn and Fe reduction [*Lyle*, 1983; *Karlin*, 1990*a*]. Apparently, paleoredox conditions must have been (cyclically) fluctuating to preserve these colors, because in steady state redox conditions the Fe redox boundary and the magnetite enrichment zone would have migrated upward at a constant rate with sedimentation. In addition, magnetic properties of the grey/sapropelitic layer such as higher J_{TS}, higher H_{cr} (Figure 4) and higher saturating fields (Figure 3) are similar to those found for the transition zone between nitrate and iron reduction in the Bettis sediments [*Karlin*, 1990*a*]. Thus, following the conclusions of *Karlin* [1990*a*], authigenic formation of pure and ultra fine-grained magnetite is the most likely mechanism for the observed rock magnetic changes. A change to a lower sedimentation rate during deposition of the overlying white marl, with less organic input, will result in a shift in paleoredox conditions. similar to trapping Mn oxides and organic carbon [*Finney et al.*, 1988], these nonsteady state conditions can promote preservation of the authigenic magnetite [*Karlin*, 1990*a*]. Clearly, additional geochemical data would be valuable to substantiate the presence of buried paleoredox indicators.

It may be expected that authigenic magnetite formation and the resulting acquisition of a chemical remanent magnetization (CRM) or a PDRM together with cyclically fluctuating redox conditions will result in a complex mixture of various natural remanent magnetizations, acquired at various times and at various depths. Indeed, *van Hoof and Langereis* [1991] have shown that in two older reversal records, the lower Nunivak and lower Cochiti from the same marine marls, there is a relative magnetization depth lag between LT and HT component, the LT component becoming locked at shallower depth than the HT component. This relative depth lag is 10-35 cm and is restricted to the beige and white marls. In the grey marls, no relative depth lag was observed, which is similar to the results from the upper Kaena interval (Figure 9). This relative depth lag could be due to a larger grain size of the LT magnetic mineral (pyrrhotite?) and hence a shallower PDRM lock-in depth. Alternatively, this relative depth lag may be due to a time lag in CRM acquisition, by which the LT magnetic minerals grow through their critical blocking volume well before the HT magnetic mineral [magnetite). In addition to a relative depth lag, both LT and HT component show an absolute magnetization depth lag, again in the beige and white marls only - which amounts to at least 60 cm (LT) and 70-80 cm (HT) in the lower Cochiti record. This is well below the PDRM acquisition depth of ~16 cm due to early sediment consolidation, as found in similar deposition rate sediments [*deMenocal et al.*, 1990], and it favors a delayed ChRM acquisition as a more likely mechanism. We note, however, that still little is known about other factors affecting the natural variability of the lock-in process.

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A considerable absolute magnetization depth lag is also found in relation to the upper Kaena reversal. Detailed paleomagnetic sampling of the entire (reversed) Kaena subchronozone in the Punta Piccola section included a sapropelitic layer in the grey marl ~100 cm below the sapropel of the upper Kaena interval. The preliminary paleomagnetic results show that in this layer intermediate and normal polarity LT and HT components are recorded. These components represent a NRM which acquired the post upper Kaena normal polarity geomagnetic field in a small and selective interval, the sapropelitic layer, at a depth of more than 100 cm. This may be due to continued authigenic CRM acquisition in this specific depth interval, or to PDRM acquisition of already existing authigenic magnetic minerals in the not yet consolidated (still high-porosity, or high water content) sapropelitic layer.

Apparently, the complicated mechanism of NRM acquisition and especially the timing of remanence acquisition makes it difficult to derive transitonal field behavior from records as presented here. Especially with respect to details of sedimentary reversal records, such as the upper Kaena excursion, we must refrain from interpreting them as true geomagnetic field behavior. Considering the observed delayed NRM in the 100 cm lower grey/sapropelitic layer, it is likely that the same or a similar processes have played a role in the upper Kaena grey/sapropelitic layer. In that case, the recorded "excursion" may be a partial record of the actual reversal, probably smoothed by the authigenic magnetite production interval, while the transition is also recorded slightly deeper (~25 cm; see Figure 9) in the grey/sapropelitic layer, under (slightly?) different redox conditions. These conditions involve either delayed CRM acquisition due to continued growth of magnetic minerals or delayed PDRM acquisition of the already existing and preserved ultra-fine magnetite. The boundary between these two different acquisition conditions is probably located at the 16-17 cm level, at the onset of the "excursion" where the sudden change of some 40° in declination occurs. Additional evidence for this assumed boundary is given by slight but equally sudden changes in H_{cr} and J_{rs}/X_0 (Figure 4), indicating an abrupt change to less fine-grained magnetite. It may be possible that in an environment critical with respect to the balance between porosity (water content) and grain size, a slight change in grain-size determines whether PDRM or CRM acquisition dominates as the remanence lock-in process.

True Geomagnetic Field Behavior?

The complex NRM acquisition process makes it extremely difficult to make a distinction between real features of the geomagnetic field and sedimentary artifacts. As discussed above, we do not interpret the excursion to be a geomagnetic feature. The low ChRM intensities during the "grey layer reversal" show a clear minimum which is unlikely to be caused by a lithological change since upon normalization with J_{rs} the minimum remains, while on the contrary the maximum in the sapropel is eliminated by normalization with J_{rs} (Figure 9). The lower ChRM intensities could therefore be related to an actually lower intensity of the transitional geomagnetic field but could also result from the averaging of normal and reversed directions. During the following "excursion", ChRM intensities are virtually constant, and normalization by J_{rs} does not significantly alter this trend. This could be due to the averaging of a relatively large normal component and small reversed (or intermediate) component, causing no apparent lowering of the intensity, but only "excursional" directions.

An important observation is the fact that both in the "excursion" as well in the "grey layer reversal", VGPs of the HT component are to a considerable extent longitudinally confined. They are constrained to two roughly opposite longitudes: those from the "grey layer reversal" are between India and Australia, while those from the "partial reversal" pass over North America (Figure 10). This corresponds respectively to a clockwise and an anticlockwise swing in declination, a difference which could be explained by a different degree of smoothing [*Hoffman and Slade*, 1986]. The rapid alternation of the LT component VGP path between the Americas path and the Southeast Asia path is not easily explained, but switching between these two paths has also been observed in other transitions, e.g., the Blake event [*Tric et al.*, 1991*b*], the upper Olduvai [*Herrero-Bervera*, 1989] and the lower Kaena [*Linssen*, 1991]. A different degree of smoothing, e.g. due to differential CRM acquisition at sampling scale, in the mottled and gradational transition from grey to brown, could perhaps cause this apparent "rapidly fluctuating" behavior. Considering the clearly delayed NRM acquisition processes that have played a role in these sediments, we find it difficult to believe that these rapid VGP alternations between Asia and the Americas are actually due to rapid fluctuations of the transitional geomagnetic field.

Tric et al. [1991*a*] found that in many reversal records from the past 10 m.y. these same two longitudinal bands are preferred, albeit in a statistical way, and they suggest that this confinement may be due to a significant dipolar component being present during the transition. They argue that smoothing of the geomagnetic signal of widely different lithologies, sedimentation rate and geographical position is unlikely to yield very similar records. On the basis of these results, *Laj et al.* [1991] furthermore suggest that there may be a link between reversal paths, the historical magnetic field and core flow pattern, and temperature distribution in the mantle. The observation that the same reversal produces VGP paths on opposite sides of the globe, however, complicates the dipolar interpretation [*Bogue*, 1991]. Recently, *Clement* [1991] used several records from the Brunhes-Matuyama to derive a reversal model involving an equatorial octupole that can produce the antipodal VGP paths like those observed. similarly, he argues that although systematic errors may result from smoothing or biasing, the geographical grouping of transitional VGPs from widely spaced sites and from different types of paleomagnetic recorders provides a strong argument that it reflects a real feature of the geomagnetic field. He also notes that often VGPs from nearby reversal records, or even within one reversal record, fall within both longitudinally confined peaks.

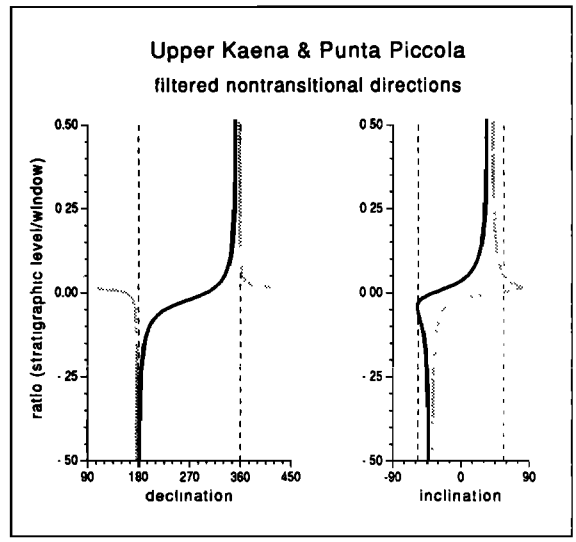
In order to test the extent to which the confinement of the upper Kaena record is determined by smoothing of nonantipodal directions, we have applied the approach followed by *Rochette* [1990]. We derive two artificially smoothed records by taking

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the average nontransitional directions from both the Punta Piccola and the upper Kaena interval (Table 2). The directions from the Punta Piccola sections are almost antipodal with respect to declination but not with respect to inclination. Smoothing results in an anticlockwise progression of the declination, and inclinations first pass through zero and become steeply normal before returning to the average value. The corresponding VGP path shows the expected and typically (near-sided) zonal behavior (Figure 12). If the reversed inclinations had been steeper than the normal inclinations, we would of course have found a far-sided VGP path. The upper Kaena stable directions are mainly nonantipodal in declination, and not surprisingly, the VGP path follows the great circle defined by those directions and passes over North America. In neither case VGPs between Asia and Australia are found.

Comparison with the actually observed VGP paths (Figure 10) shows that the "grey layer reversal" initially proceeds along an Asian path, according to both HT and LT component, in contrast to the artificial records. The HT component remains confined to this path, although a strong eastward deviation is observed. The LT path, on the other hand, starts alternating between Asia and the Americas, while the last part is mainly confined to North America. Most conspicuously, the LT component VGPs on the Americas side of the globe are exactly on the path defined by smoothing of the upper Kaena stable directions (Figures 10 and 12). In addition, all VGPs from the "excursion" are located in North America, both from the HT component (Figure 10) and from the LT component (not shown in Figure 10 for readability, but see Figure 8). Hence it seems as if initially the reversal in the grey layer is a true recording of a path over Asia but becomes increasingly confined to a path over the Americas which is the same as that determined by smoothing of nontransitional directions from the upper Kaena interval. The following "excursion" is then possibly only a partial and in addition strongly smoothed record of the same reversal, since its VGPs are only confined to North America.

Finally, an earlier interpretation of the preceeding lower Kaena and both Mammoth reversals [*Linssen*, 1991] suggested that the major directional changes of these three reversals take place in 5-10 cm, or in 1-2 kyr. The intervals of minimum intensities are, on the contrary, very similar (20-30 cm or 4-6 kyr) to the interval found in the upper Kaena record. The character of the complete Mammoth and Kaena reversal succession seems rather complex: the lower Mammoth is recorded as a single reversal, the upper Mammoth is followed by an excursion in inclination, the lower Kaena is followed by a full excursion to normal polarities and high intensities, and the upper Kaena again is followed by an excursion. The VGP paths of the older reversals are generally well confined to great circles: the one from the lower Mammoth passes over India and those from the upper Mammoth and lower Kaena pass roughly over North and South America, while the R-N path of the full excursion follows a path over India again [*Linssen*, 1991]. VGPs of the upper Kaena are confined to a path between India and Australia (HT component) or alternate between this path and a path over the Americas (LT component), while the VGPs of the excursion are confined to North America. However, the fact that these earlier interpreted



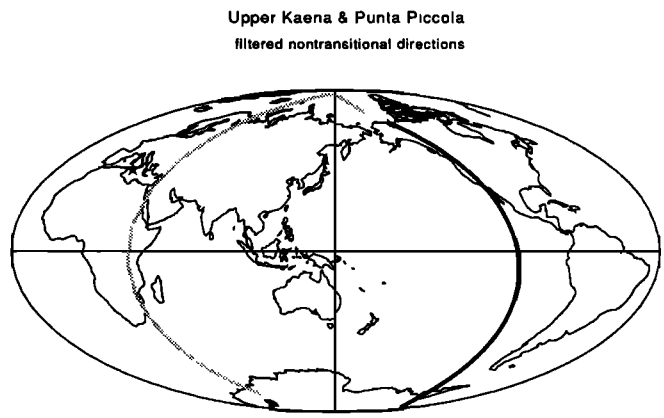


Fig. 12. Filtering of the mean pretransitional and posttransitional directions of Table 2, from the Punta Piccola section (shaded line) and from the upper Kaena interval (solid line), using the approach of *Rochette* [1990]. The reversal occurs - instantaneously - at level 0.00; ordinate gives the fraction of the smoothing window (minus 0.5) that "sees" the normal polarity. The corresponding VGPs on the Aitoff projection show a typically zonal and near-sided transition (Punta Piccola) and a path over western North America (upper Kaena). The final part of the latter coincides with the final part of the LT component VGP path (Figure 10).

records have been established from the same sediments, with the same cyclically fluctuating lithology, warrants a reinterpretation in terms of sedimentary artifacts rather than in terms of true geomagnetic field behavior.

CONCLUSIONS

A detailed paleomagnetic record of the upper Kaena (R-N) polarity reversal from southern Sicily shows that two characteristic components can be determined separately: a low-temperature (LT) component and a high-temperature (HT) component. Both components have been acquired in an early stage following deposition and are probably carried by

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authigenically formed and microbially mediated magnetic minerals in a suboxic environment.

A conventional interpretation of the record would suggest a transition which takes places in approximately 3.3 kyr, with lowered intensities during 5.6 kyr. After a stable period of 3.3 kyr, it is followed by a significant "excursion" during 3.6 kyr.

However, based on the lithology, the rock magnetic and paleomagnetic properties, as well as on earlier results from the same marls [*van Hoof and Langereis*, 1991], we conclude that the features observed in the upper Kaena record, and also those in the three preceding records [*Linssen, 1991*], are largely determined by the remanence acquisition process in these sediments. Most likely, the remanence has been smoothed by CRM and/or PDRM acquisition, at least to the extent that finer details such as secular variation and possible rapid geomagnetic fluctuations have been lost from the record. More importantly, we conclude that cyclically fluctuating paleoredox conditions have resulted, first in a delayed paleomagnetic record in a carbonate-poor (and possibly higher-sedimentation rate) grey/sapropelitic layer due to magnetite preservation and second in an apparently posttransitional "excursion" which we believe is a partial, strongly smoothed but less delayed record of the same [upper Kaena) reversal.

It is remarkable that the observed VGP paths are confined either to a longitude between India and Australia, or to a longitude over South and North America. These two geographically confined paths were recently suggested to be related to the reversal process [*Tric et al.*, 1991*a*; *Laj et al.*, 1991; *Clement*, 1991]. The longitudinal confinement of the high temperature component to a path over India/Australia may be a real feature of the geomagnetic field. The major confinement of the LT component and of the "excursional" HT component to a path over the Americas agrees well with the path derived from smoothing the nonantipodal stable directions from the upper Kaena record.

This study emphasizes the importance of including detailed paleomagnetic/rock magnetic, lithological/geochemical data in studies of sedimentary reversal records in order to determine the reliability of the observed "transitional" features. We note that numerous "geomagnetic" features like longitudinal confinement, excursions/rebounds and rapid fluctuations from sedimentary records have been reported in literature which, according to the results of our study, could be due to local biasing effects.

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